

UNIT – I INTRODUCTION TO JAVA**9L, 12P**

Introduction to Java – JVM - Data Types, Variables, Operators, Expressions – Control flow Statements – Methods - Arrays – Classes and Objects –Constructors - Access Specifiers –Static members- this –constants- - String Class- Working with Data and Time API

PRACTICALS:

1. Develop programs using Java Basic Constructs and Arrays using any standard IDE like NETBEANS / ECLIPSE
2. Programs to illustrate concept of class and static classes and methods
3. Implement programs using string class, Date and Time API

UNIT – II POLYMORPHISM AND INHERITANCE**9L, 12P**

Overloading Methods – Static, Nested and Inner Classes. Inheritance– super Classes and sub classes -Overriding –Downcasting – Operator instance of - Abstract Classes – final classes - Packages – Interfaces.

PRACTICALS:

1. Implement abstract classes, overloading and overriding
2. Implement Interfaces

UNIT – III EXCEPTION HANDLING AND MULTITHREADING**8L, 12P**

Exception Handling – Java's Built-in Exceptions – User defined Exception - Assertions. Multithreading- Priorities–Synchronization – Avoiding Deadlocks– Wrappers– Autoboxing and Unboxing.

PRACTICALS:

1. Programs using Exception handling
2. Programs using Multi-Threading and synchronisation

UNIT – IV FILE STREAMS AND DATABASE**9L, 12P**

Java I/O- Reading and Writing Files - Regular Expressions – Streams API - Object Serialization – Generic collections – Generic Classes - Lists, set, Maps – GenericMethods - Lambda expressions – Databases with JDBC.

PRACTICALS:

1. Programs using Streams API and File I/O for reading and writing the contents in sequential and random order.
2. Programs using Generics classes and methods
3. Event-driven programs for GUI applications with Database Connectivity

UNIT – V WEB DEVELOPMENT AND FRAMEWORKS**10L, 12P**

Event handling: Event, Listeners and adapter classes, anonymous inner classes. Abstract Windowing Toolkit (AWT): Button, Label, Checkbox, Checkbox Group, Text Field, Text Area, Choice, List, Menu, Panel, Scrollbar and Swing components, Layout managers, Complex Components - Java Servlets - Apache Tomcat –Java Web Framework -Spring Boot

PRACTICALS:

1. Creation of servlet based applications
2. Design a Java Web application using Spring Boot

TOTAL: 45L + 60P = 105 PERIODS**COURSE OUTCOMES:**

Upon completion of the course, the students will be able to

1. Construct programs using Object Oriented Design principles like encapsulation, abstraction, polymorphism, inheritance and typing. (Apply)
2. Develop applications with handlers for user-defined exceptions, according to the given requirements (Apply)
3. Construct efficient multi-threaded programs with synchronization constructs. (Apply)
4. Develop interactive GUI applications with event handling that provide rich user experience. (Apply)
5. Construct programs using the suitable Collection classes and interfaces for efficient modelling of the objects and entities of the program. (Apply)
6. Develop Web Applications that use file input and output using any framework (Apply)

REFERENCES:

1. Y. Daniel Liang, "Introduction to Java Programming and Data Structures, Comprehensive Version", 12th Edition, Pearson Education, 2024.
2. Paul Dietel and Harvey Deitel, "Java - How to Program Early Objects", 11th Edition, Pearson Education, 2018.
3. Craig Walls, "Spring in Action", 3rd edition, Manning Publications, 2011.
4. Herbert Schildt, "Java: The Complete Reference", 11th Edition, McGraw-Hill Education, 2018.
5. Sachin Malhotra, Sourabh Choudhary, "Programming in Java", Revised 2nd Edition, Oxford University Press, 2018.
6. Cay S. Horstmann, "Core Java - Vol. 1, Fundamentals", 11th Edition, Pearson Education, 2018.
7. <https://spring.io/projects/spring-boot>

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	3	2	1	3	1	-	2	1	2	3	3	3	2
CO2	3	2	3	2	2	3	1	-	2	1	2	2	3	1	3
CO3	3	1	3	2	2	3	1	-	2	1	2	3	3	1	1
CO4	3	1	3	3	2	3	1	-	2	1	2	3	3	3	3
CO5	3	1	3	3	1	3	1	-	3	1	2	3	3	3	2

1-low,2-medium,3-high,'-'no correlation